

- Q.27 Write the advantages of heat treatment. (CO1)
 Q.28 Differentiate between slip and twinning. (CO2)
 Q.29 Explain lead bath furnace. (CO2)
 Q.30 What austenite. (CO2)
 Q.31 Write the characteristics of carburizing. (CO3)
 Q.32 Write the uses of nitriding. (any five) (CO3)
 Q.33 Write purpose of tempering. (CO4)
 Q.34 What is martensite? (CO5)
 Q.35 Write any five application of heat treatment. (CO6)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What do you mean by heat treatment furnace? What are its different types? Explain any two. (CO1)
 Q.37 Explain the electric furnace with neat sketch. (CO4)
 Q.38 Explain T.T.T. diagrams. (CO5)

Note : Course Outcome (CO) mentioned in the questions paper is for official purpose only.

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5th Sem / T&D, Found & Forg. Sub : Heat Treatment

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is the hardest constituent of steel? (CO1)
 a) Ledeburite b) Austenite
 c) Bainite d) Martensite
- Q.2 For steel, which one of the following properties can be enhanced upon annealing? (CO1)
 a) Hardness b) Toughness
 c) Ductility d) Resilience
- Q.3 In Annealing, cooling is done in which of the following medium? (CO2)
 a) Air b) Water
 c) Oil d) Furnace
- Q.4 In normalizing, cooling is done in which of the following medium? (CO2)
 a) Air b) Water
 c) Oil d) Furnace

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Q.5 Mild steel can be converted into high carbon steel by which of the following heat treatment process? (CO3)

- a) Annealing b) Normalizing
- c) Case hardening d) Nitriding

Q.6 Upon annealing, eutectoid steel converts to which of the following? (CO3)

- a) Pearlite b) Cementite
- c) Austenite d) Martensite

Q.7 Alloy of copper and zinc is known as _____. (CO3)

- a) Brass b) Bronze
- c) Duralumin d) Nichrome

Q.8 Alloy of Ni and Fe is termed as _____. (CO4)

- a) Brass b) Bronze
- c) Duralumin d) Invar

Q.9 Which of the following alloy is widely used in thermocouples? (CO4)

- a) Brass b) Bronze
- c) Duralumin d) Nichrome

Q.10 Major constituent of Duralumin alloy is _____. (CO5)

- a) Copper b) Nickel
- c) Iron d) Aluminium

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The object of annealing is _____. (CO1)

Q.12 Nitriding is a process for _____. (CO1)

Q.13 Name any two semi-conductors. (CO1)

Q.14 Define heat treatment. (CO2)

Q.15 Define tempering. (CO2)

Q.16 Name two types of annealing. (CO2)

Q.17 Pearlite is a combination of _____. (CO3)

Q.18 Name the purest form of Iron. (CO3)

Q.19 Give the full form of T.T.T. (CO5)

Q.20 Define hardening. (CO6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Define Ferrous metals with one example. (CO1)

Q.22 Define creep. (CO1)

Q.23 Define ductility of a material. (CO1)

Q.24 Name four types of annealing. (CO2)

Q.25 Define process of annealing. (CO2)

Q.26 Differentiate between annealing and hardening. (CO1)